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OPTIMUS PRIME CONSULTING

Building Your Better Future

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WHAT WE SOLVE

Many organizations face a recurring challenge during their planning cycles: They have far more proposed initiatives than they have the budget, people, or resources to execute. Strategy, product, IT, operations, and marketing teams all submit projects that can be important, but leadership can only fund a subset of them. In practice, portfolio decisions are often driven by internal politics rather than a structured view of how to allocate limited resources for maximum value. It becomes difficult to answer seemingly basic questions like “Which combination of projects creates the most value under these constraints?”. Or “What should be cut if the budget is reduced by 10%?”

This is inherently an optimization problem: it involves allocating multiple constrained resources across a portfolio of projects with different costs, benefits, risks and dependencies. The complexity grows quickly as projects draw on different resources, have prerequisites or mutually exclusive options, and serve competing stakeholder priorities. Without a structured, rigorous approach, organizations underutilize their resources, end up with a funded portfolio that does not align with the stated strategy, and create an environment where teams often perceive decisions as politically motivated rather than genuinely optimal.

Our service is designed to address this kind of resource allocation problem by framing project portfolio selection explicitly as a constrained optimization problem and providing a repeatable way to evaluate trade offs and different scenarios.

WHO ARE WE FOR?

The primary clients for this service are mid to large sized organizations that manage a significant pipeline of initiatives during their planning cycles and face real budget or capacity constraints. This includes technology and software companies, manufacturers with ongoing product development, financial services firms, healthcare systems, and other service enterprises in which projects compete for limited funding and specialized talent.

Within these organizations, the key stakeholders are strategy and corporate development teams, PMOs (program management offices), finance and budgeting teams, and technology or operational leaders. They are responsible for translating strategy into a feasible set of funded projects, making trade offs and justifying decisions to leadership and their teams. Our ideal clients are those who are managing a large portfolio of projects per cycle, experiencing binding budget/headcount constraints, and are looking for a more transparent and data-driven approach. For these organizations, our portfolio optimization service, which uses an explicit optimization model to propose funding plans and scenarios, can materially improve decision quality, alignment, and trust.

OUR CAPABILITIES

Optimus Prime Consulting builds your better future with you, for you, by optimizing your budget, overall spending, and protecting client and investor relationships, given your unique constraints.

We formulate your problem into a mathematically solvable model, implement it in Excel or OR-Tools, depending on the programming type and complexity, and compile it into a straightforward report and presentation with our recommendations.

Our modeling approach, fundamentally, runs on a structured method of building a model that runs (aligning on the problem, identifying decision variables, formulating the objective function, and defining the constraints), iterating until feasible, and refining for quality/validation of an optimal solution.

This approach allows scalability to larger problems by systematically adding complexity until feasibility is reached, then refining the solution until further improvements are deemed unworthy of the time or resources needed. We are dedicated to providing results that will minimize your costs while maximizing your profit and growth. Optimus Prime Consulting builds your better future with you, for you, by optimizing your budget, overall spending, and protecting client and investor relationships, given your unique constraints.

OUR METHODS

Our go-to tools for tackling problems are mixed-integer programming, constraint programming, and A/B testing. These three methods allow us to cover the majority of cases quickly and efficiently, and in the cases that they don't give us the feedback we need to pivot how we serve the customer. In all cases, we require our customers to provide their requirements and their business goals so that we can align our work with their needs.

Having the tools for the job is only half the story, using them is what's key, and we our customers the trouble of spending valuable resources and development time on learning how to leverage each tool to their maximum potential.

We always start at the human scale: business problems are people problems first and foremost. Who needs what? When do they need it? What's the holdup? These are all seemingly simple questions that get asked every day in a business, but the devil is in the details. By getting an idea of how things should work for each stakeholder, we can combine each perspective and create a holistic optimization goal that maximizes true business value. This is the foundation of our capabilities, and allows our team to apply each technical tool efficiently and with precision to the problem at hand.

WHY OPTIMUS?

Has the same answer as 'why optimize?' It can be difficult to justify spending money on optimizing business operations; after all, the business is running the way things are already, but not adapting to the times is an opportunity cost in and of itself. It costs money in future profits to stagnate, and business processes can hinder growth when they don't adapt to meet changing market conditions.

This is where Optimus Prime Consulting comes in: immediate relief for the bottom line and ongoing support to increase the lifetime value of our relationship. We understand that our customers need their problems solved as soon as possible, but we also know that the best solution takes time to find and verify, and that's time our customers often don't have. We do the tedious iteration and improvement that characterizes optimization, providing immediate access to profit while working to maximize gains into the future as well.



**CASE
STUDIES**

ORANGE ELECTRONICS

Key Problem

On October 3rd, a flood in Taiwan severely damaged the Kaoshiung facility. The Kaoshiung facility typically produces 40% of Orange Electronics' global output, but it was reduced to 35% capacity. Two critical delivery windows need to be met in the coming weeks using the Kaoshiung, Chennai, and Guadalajara facilities. There are also constraints on materials, time to manufacture, etc.

Problem Understanding

Production of each device type needs to be allocated across the three facilities to fulfill both delivery weeks on time. The costs of express shipping and overtime need to be within a \$15 million emergency budget, and the total production, shipping, and penalty costs must be minimized. If goals are not met, client relationships are at risk.

Results & Impact

The total cost is \$271,319,811, with \$259,896,000 coming from the bottom line and \$11,423,811 from the emergency budget to cover express shipping costs.

Both the Week 45 and Week 46 delivery requirements were met, emergency budget usage was less than allocated, and client relationships were preserved.

Methodology

The data provided in the Excel spreadsheets and stakeholder interviews needed to be cleaned to remove inconsistencies in naming conventions, formatting issues, and missing values. After data cleaning was completed, we formulated the model by setting up workbook tabs for the parameters, decision variables, constraints, objective, and solution sheet. We used an iterative approach to develop a feasible solution and validated it by testing the Excel solver results against similar solutions from previous runs. This was to confirm that the result was within the expected margin of error and was acceptable to issue as a recommendation.

Recommendations

For Week 45, produce 70,000 phones at Chennai, 3094 tablets at Kaoshiung, 21,906 tablets at Chennai, 1 laptop at Chennai, and 47,999 laptops at Guadalajara. Use standard shipping to deliver all devices for Week 45.

For Week 46, produce 165,000 phones at Chennai, 66,906 tablets at Kaoshiung, 28,094 tablets at Chennai, 15,000 laptops at Kaoshiung, 12,999 laptops at Chennai, and 22,001 laptops at Guadalajara. Use standard shipping for all Kaoshiung and Guadalajara production, and express shipping for Chennai.

ADSTREAM

Key Problem

AdStream's ad serving system is losing advertisers due to poor performance. Their current system allocates impression inventory inefficiently using greedy heuristics, which is simple and fast, but can overlook more optimal solutions. The revenue is below potential, and there is significant revenue loss from brand safety violations that trigger automatic 20% discounts for advertisers. Advertisers have also been complaining about poor topic matching and appearing alongside competitors, which has led to multiple threatening to cancel.

Problem Understanding

Since revenue is low and advertisers are unhappy, AdStream needs to optimize its system to retain clients and demonstrate profitability to investors.

Methodology

When formulating how and what to optimize, the business requirements taken into consideration were maximizing total revenue, respecting website capacity limits, enforcing advertiser min/max impression requirements, ensuring that each active advertiser appears on greater than or equal to the minimum number of websites needed, enforcing minimum placement size, and only allocating to topic-matched websites.

Specifically when looking at the heuristic system for allocation and assignment, the areas of improvement were impression placement and gross revenue. To implement these changes using OR-Tools with a MILP model, it was necessary to separate the advertiser data set into two (advertiser maximums and advertiser minimums), find each advertiser's most valuable topic, generate valid (advertiser, website) pairs for allocation, and initialize the solver and solve.

Moving on to the heuristic system for advertiser ordering, the main issue was the violation rate on websites. The business requirements were that competitors must be greater than or equal to 2 positions apart in the sequence, a valid ordering that satisfies spacing constraints must be found or if no valid ordering exists for a website, the original heuristic approach is used, and each website is solved independently. To implement the optimized ordering, a local search algorithm needed to group placements by website, generate a map of all the competitors, generate initial orderings for each website by checking if the website has existing violations and moving advertiser orderings to reduce violations, and update advertiser orderings for each website.

Finally, we validated our solutions using the snowflake data.

Recommendations

Deploy the optimized two-stage model as the default engine since it uses 100% of available impressions (1933.7K / 1933.7K) and lifts gross revenue from \$96.4K to \$110.6K and net revenue from \$78.5K to \$90.0K (+14.6%).

Results & Impact

AdStream was able to increase its gross and net revenue, while still using 100% of available impressions.

Using the optimized system, AdStream can benefit from an increase in revenue (even more so if scaled up due to more ordering options) and improve client relationships.